ANSHIKA SAHU

Generative AI & LLM Engineering | RAG Systems | GPU Inference | Neuroscience Data Science

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PROFESSIONAL SUMMARY

Neuroscience student and GenAI developer who builds, benchmarks and deploys LLM systems end to end. Built a **neuro-behavioral analytics platform** (6 mice, 105 sessions, 30K+ trials) with a **hybrid-search RAG assistant** (BM25 + dense vector retrieval, cross-encoder reranking), **benchmarked LLM inference from limited-memory RTX GPUs up to NVIDIA H100s on RunPod**, and deployed the full backend on a self-provisioned **GCP VM** with Docker. Brings healthcare insight from **1,000+ hours as a clinical medical assistant**, 400+ volunteer hours and 500+ research hours.

TECHNICAL SKILLS

Generative AI & RAG: Retrieval-Augmented Generation, hybrid search (BM25 lexical + dense vector), Reciprocal Rank Fusion, cross-encoder reranking, HyDE, Self-RAG & Corrective RAG (CRAG), prompt & system-prompt engineering, prompt-injection mitigation, LLM response evaluation

LLM Serving & GPU Compute: Ollama (Llama 3.2, Gemma 3, Qwen), RunPod, NVIDIA H100 & RTX GPUs, streaming inference, latency/throughput benchmarking, model hot-swap & memory tuning

Cloud & MLOps: Google Cloud (Compute Engine VM), RunPod, Docker, Docker Compose, Linux/WSL2, Streamlit Cloud, ngrok, Git/GitHub, automation scripts

Programming & Data: Python (pandas, NumPy, SciPy), Bash, PowerShell, MATLAB, JSON/YAML; ETL pipelines, schema standardization, Parquet/PyArrow, data validation

Neuroscience & Visualization: Signal detection theory (d', criterion), Go/No-Go paradigms, widefield calcium imaging (ΔF/F), time-series analysis; Streamlit, Plotly

GENAI & DATA SCIENCE PROJECT

Code: github.com/anshikasahu2005/Mouse-Data-Analysis

Multi-Mouse Neuro-Behavioral Analytics Platform | *Python · Ollama · BM25 · Cross-Encoder · RunPod H100 · GCP · Docker · Streamlit*

- Built a **reusable ETL pipeline** converting raw MATLAB data to Parquet/CSV: **6 subjects, 105 sessions, 30,128 trials, 127K lick events, 663K frame timestamps**.
- Implemented **signal-detection scoring** (Hit/Miss/FA/CR) with **d' and criterion** across animals; processed calcium-imaging **ΔF/F** into **131 peak-activity cortical maps**; new subjects onboard with **zero code changes**.
- Built a RAG assistant that grounds answers in **live experiment statistics** (hit/FA rates, d') through a domain-specific system prompt.
- Implemented **hybrid retrieval** combining **BM25 lexical search with dense vector search**, fused with **Reciprocal Rank Fusion** and re-ranked by a **cross-encoder** to surface the most relevant context.
- Added **HyDE** query expansion, a **Corrective RAG (CRAG)** step that grades retrieved context and re-queries on low relevance, **prompt-injection guards** and automated answer evaluation.
- Diagnosed a **small-model reasoning failure** (3B model mis-ranked subjects); fixed by pre-ranking retrieved aggregates, **improving factual accuracy** without a larger model.
- Benchmarked the same LLM stack across compute tiers, from **limited-memory consumer RTX GPUs up to NVIDIA H100s on RunPod**, comparing latency (**~5 s vs. ~150 s**), throughput and memory use to identify the best performance-per-cost setup.
- Added **model hot-swap, warm-up and memory limits** so one codebase runs on constrained RTX hardware and scales to H100 cloud GPUs with **streaming responses**.
- Provisioned a **GCP Compute Engine VM from scratch** (Linux setup, firewall rules, SSH) and deployed the **entire backend** (LLM service, RAG pipeline, dashboard) on it.
- Containerized with **Docker Compose** (non-root, health checks); cut build context from **9.3 GB to 88 MB**; also deployed on Streamlit Cloud.

RESEARCH, CLINICAL & VOLUNTEER EXPERIENCE

Research Assistant | *Zagha Lab, UC Riverside* **09/2024 – Present**

- Cortical sensory-detection research: behavioral training, widefield Ca^{2+} imaging and **MATLAB-based behavioral data analysis** (561 hours).

Clinical Medical Assistant | *Silicon Valley Sports Med. · City Health Urgent Care* **2023 – 2025**

- **1,000+ hours of clinical experience**: roomed 30–40 patients daily; vitals, injections, wound care, EKGs, charting.

Community Health Volunteer | *COPE Health, Street Medicine, UCR Probe Scholar* **2024 – Present**

- **400+ volunteer hours** supporting under-resourced patients with care and resource navigation.

EDUCATION & CERTIFICATIONS

B.S. Neuroscience, Minor in Data Science | *University of California, Riverside* **2023 – 2027**

Certifications: Clinical Medical Assistant, UEI-U.S. Colleges (2023) | NHA Certified Medical Assistant | CPR/BLS